

EEG Documentation

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1 Abstract

While participants perform cognitive tasks, simultaneous electroencephalography (EEG) recordings enable us to examine abstract representations in the human brain. We are specifically looking into decoding non-sensory statistical information—such as the switch likelihood associated with an image—which could be strategically used to enhance cognitive flexibility and reduce the behavioral costs of task-switching. Moving beyond traditional EEG analysis that examines activity from individual electrodes across time, we leverage support vector machine (SVM) models to decode patterns from multiple electrodes. By training this model separately at different time points and employing cluster-based permutation tests, we are able to show that decoding accuracy peaks approximately 100 ms to 200 ms after stimulus onset—much earlier than what traditional methods have identified. Furthermore, by examining feature importance weights, we identify areas of the brain that play a key role in representing switch likelihood information. Lastly, we also find cross-subject generalization, indicating that multivariate EEG patterns reliably capture abstract representations at the population level. This project lays the groundwork for understanding how humans utilize probabilistic information to enhance cognitive flexibility.

2 Context

Bin 1: False frequent switch cue, preceded by a correct trial, current trial is also correct

Bin 2: False frequent repeat cue, preceded by a correct trial

Bin 3: Frequent-switch item, current trial correct, preceded by a correct trial

Bin 4: Frequent-repeat item, current trial correct, preceded by a correct trial

This study is focusing on bin 3 and bin 4 in order to determine where and how early identification of the rule can be detected.

3 Preprocess

1. Set sample rate to 250 samples per second
2. Set frequency cutoffs to 0.2 – 20 Hz

3. Twice - Detect eye blinks and eye saccades
 - (a) Run ICA with sample rate of 100 samples per second
 - (b) Manually select components that resemble eye blinks or eye saccades
 - (c) Subtract the manually selected components
4. Set reference point to Cz
5. Detect bad channels based on normalized kurtosis exceeding 5
6. Interpolate bad channels using “spherical” method
7. Use “pop_artmwpbth” with a threshold of 100 Hz, window size of 200 ms, and window step size of 100 ms

4 Bin Selection

Subjects with less than 70 bin 3 or bin 4 conditions were excluded in further analysis.

Total subjects: 79

Total subjects rejected: 22

Average percent of trials rejected per subject: $\sim 29\%$

5 Significant Clusters

Traditionally, conducting a statistical test on a high number of sensor and time pairs would have a high false positive rate. We used cluster-based permutation tests in order to reduce the false positive rate due to the high temporal resolution and number of electrodes. However, using cluster-based statistical tests can not be used to justify precise location and timing, therefore this should be considered when looking at the results.

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1: function CLUSTER PERMUTATION TEST(condition 1, condition 2)
2:   Select samples where t value is greater than some threshold
3:   Cluster samples based on some criteria
4:   Calculate cluster-level statistic (sum of t values)
5:   Repeat lines 2 to 4 with shuffled labels  $x$  times.
6:   Calculate how likely the original clusters were compared with the shuffled
    labels clusters
7: end function

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5.1 Multi-subject analysis

To conduct a multi-subject cluster-based permutation test, each subject averaged all of their trials with both respective conditions. The shuffling step will randomly swap the subject-specific averages within each subject independently.

5.1.1 Temporal

The criteria to cluster samples for the cluster-based permutation test performed per electrode and is only based on time. Continuous times are clustered if they are both past the threshold.

5.1.2 Temporal and Spatial

The criteria to cluster samples for the cluster-based permutation test is based on time and spatial adjacency. Continuous times are clustered if all adjacent electrodes are also past the threshold.

Results:

Total clusters: 30 (16 positive, 14 negative)

Total positive clusters p-value less than 0.025: 0

Total negative clusters p-value less than 0.025: 0

6 SVM

The SVM model was trained to determine if a trial belongs to bin 3 or bin 4 using 64 electrodes. For individual trials, the SVM used balanced penalties for classes because of the imbalance between bins. We used 10 k-folds to achieve more accurate metrics. SVMs were trained and tested per time steps (0.004s).

6.1 Individual Subjects

This section attempted to train an SVM per each individual using their raw trials. There were two methods to train the SVM per subject: Individual trials and averaged trials.

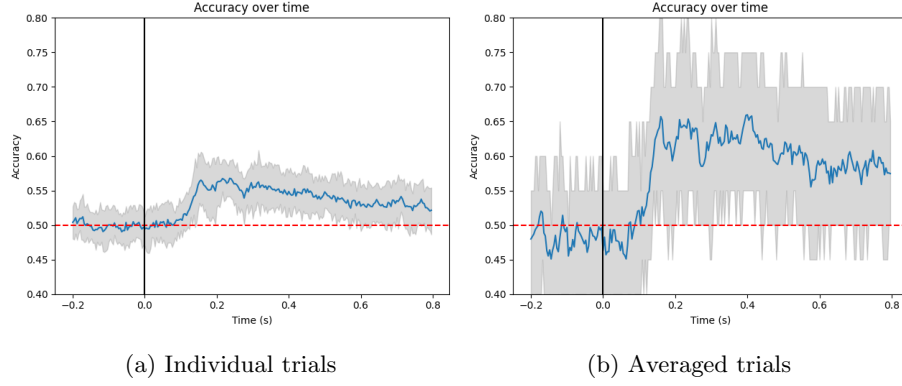


Figure 1: Average accuracy per SVM. The 25th and 75th percentiles are shaded in gray. Random chance (0.5) is marked with a horizontal dashed red line.

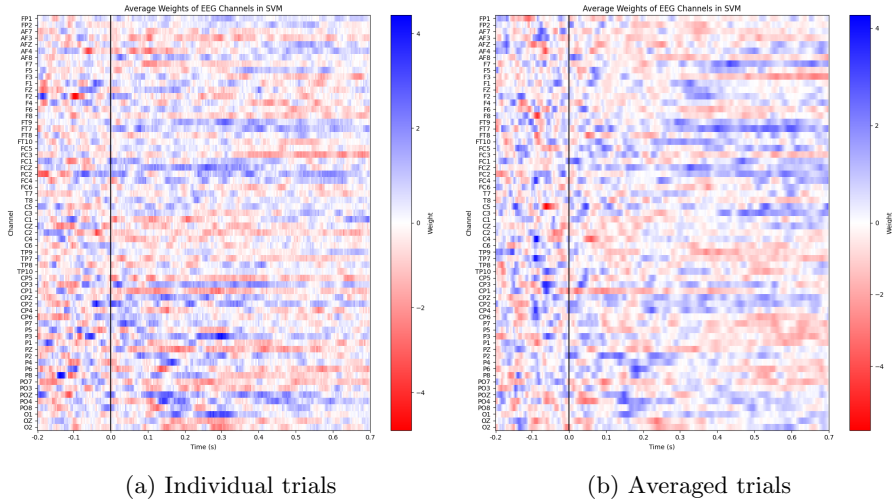


Figure 2: Average weights per SVM across time. Stimulus onset (0 ms) is marked with a black vertical bar. The color bars are determined on minimum and maximum weight values.

The average weights in SVMs across time show how important electrodes are relative to each other across time as shown in Figure 2. Before stimulus onset (0 ms), the weights vary a large amount. The average weights' signs per SVM are mostly consistent after stimulus onset between individual trials and average trials, suggesting that some electrodes are consistency important across time and tests. Additionally, the individual trials weights have few electrode weights with higher importance weights whereas the averaged trials have more balanced lower importance weights across electrodes. This suggests that individual trial

decoding relies more on a specific subset of electrodes, and averaged trials are able to balance between electrode weights to achieve a higher accuracy.

6.1.1 Statistically Significant Electrodes

The above experiment was repeated with a modified procedure for training to exclude irrelevant electrodes. To get an unbiased selection of electrodes, one third of the training fold was used to get statistically significant electrodes, and the other two thirds were used to train the SVM.

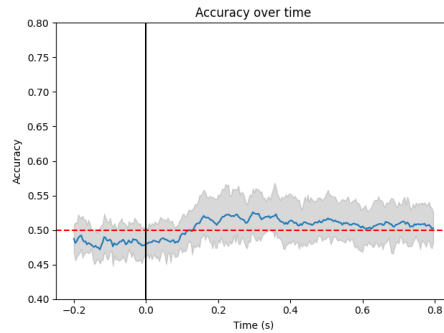


Figure 3: Average accuracy per SVM. The 25th and 75th percentiles are shaded in gray. Random chance (0.5) is marked with a horizontal dashed red line.

With this procedure, the accuracy did lower, but it was also able to show above chance accuracy after around 100 ms.

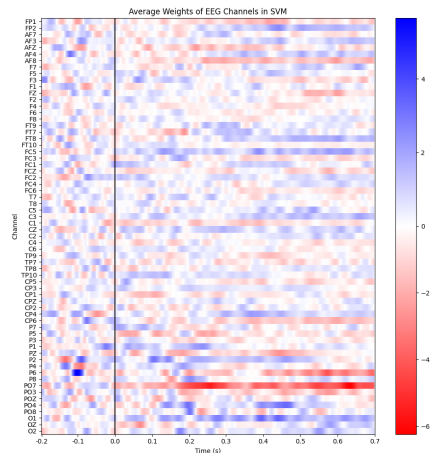


Figure 4: Average weights per SVM across time. Stimulus onset (0 ms) is marked with a black vertical bar. The color bars are determined on minimum and maximum weight values.

The SVM's across folds mainly focus on a few electrodes, possibly useful for generalization across subjects.

6.2 Multi-subject analysis

51 subjects ($\sim 90\%$) were used to train an SVM per time step and 6 subjects ($\sim 10\%$) were used to test the accuracy. This was repeated 10 times per time step to provide more accurate metrics.

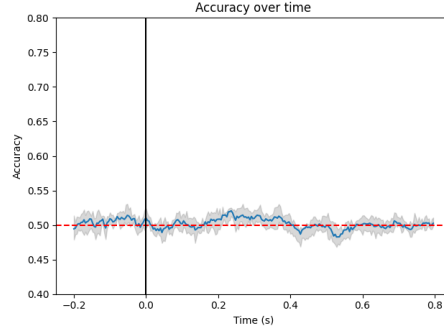


Figure 5: Average accuracy per SVM. The 25th and 75th percentiles are shaded in gray. Random chance (0.5) is marked with a horizontal dashed red line.

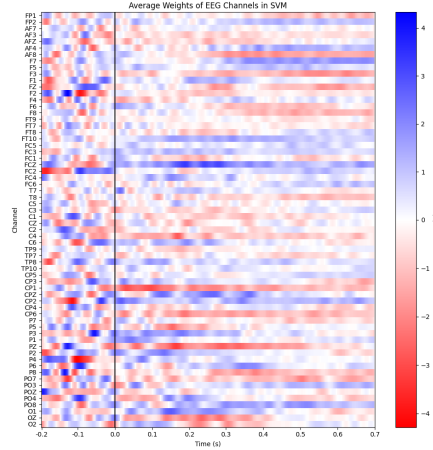


Figure 6: Average weights per SVM across time. Stimulus onset (0 ms) is marked with a black vertical bar. The color bars are determined on minimum and maximum weight values.

The accuracy is reduced to just barely above chance during some time ranges. However, the weights of the SVM remain steady across time, possibly indicating

that there is some general item-specific representation.

7 Deep Learning

Due to low accuracies from both individuals and across subjects, deep learning methods were applied as they are better able to capture patterns than traditional methods. Using Braindecode (<https://braindecode.org>, <https://github.com/braindecode/braindecode>), an open-source Python library focused on using deep learning models to decode raw electrophysiological brain data, we were able to create a few procedures to decode our data.

7.1 Shallow ConvNet

The Shallow ConvNet model was created based on the paper, <https://onlinelibrary.wiley.com/doi/10.1002/hbm.23730>, and aims to decode band power features from raw EEG data. Shallow ConvNet contains 104,082 learnable parameters. Shallow ConvNet contains temporal convolutions and spatial filters.

On our dataset, we found that setting the learning rate to 0.0001, weight decay to 0.05, and batch size to 64 for both individual and cross-subject tests lead to more smoother accuracies across epochs. We used 80% to train and 20% for validation. Subject 2 has 268 trials, and across subjects, there are 17,421 total trials.

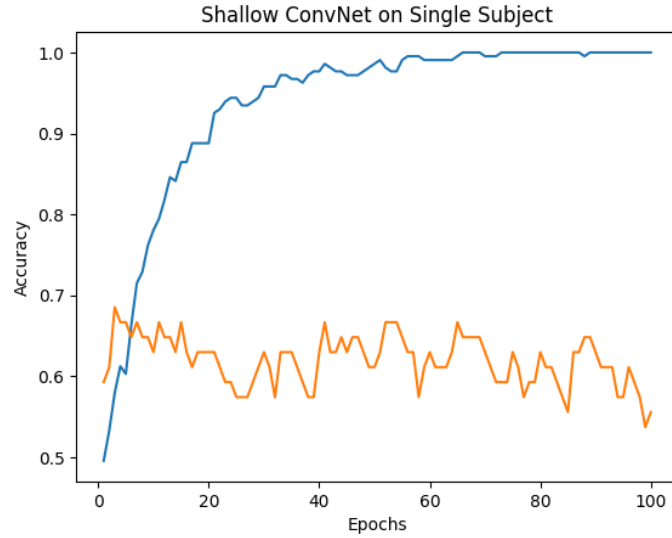


Figure 7: Accuracy across time for Shallow ConvNet only on Subject #2.

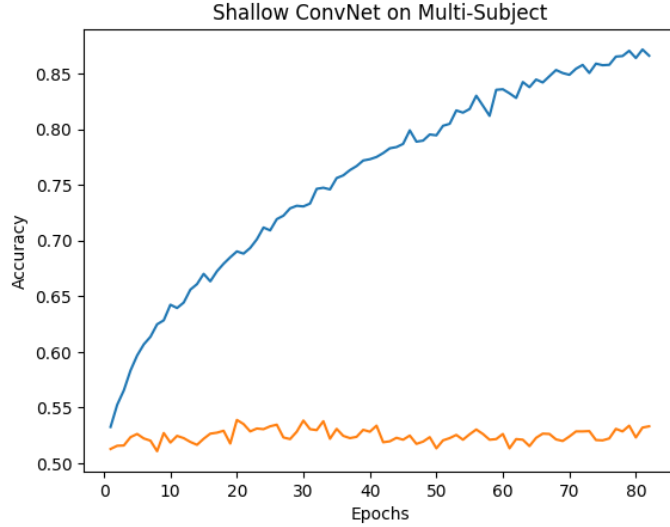


Figure 8: Accuracy across time for Shallow ConvNet across all subjects.

The model easily overfit for both cases, possibly pointing to the model learning noise specific to each trial instead of actual patterns. Additionally, the number of parameters is almost 10 times the number of total trials, also leading towards memorization instead of generalization from the model. The rest of the experiments will use an early stopping method of stopping after 15 epochs if the validation doesn't increase.

7.1.1 Fewer Parameters

To mitigate overfitting, the number of parameters can be reduced. Specifically, we changed the number of temporal filters from 40 to 20, length of the temporal filter from 25 to 15, and the number of spatial filters from 40 to 20. This lead the model to go from 104,082 learnable parameters to 26,402.

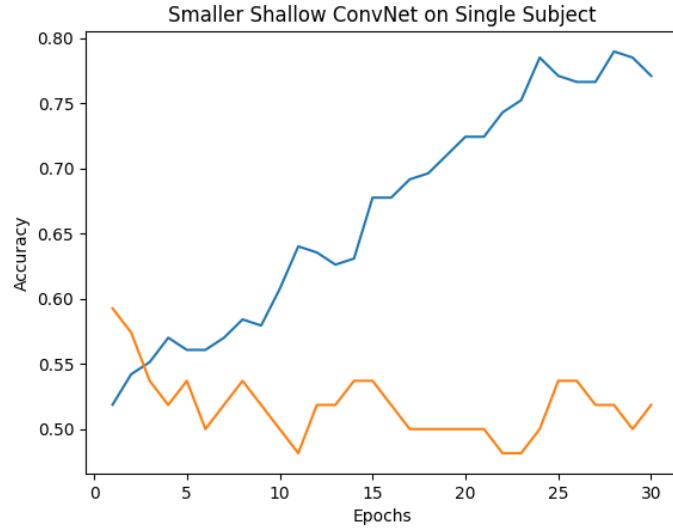


Figure 9: Accuracy across time for Smaller Shallow ConvNet only on Subject #2.

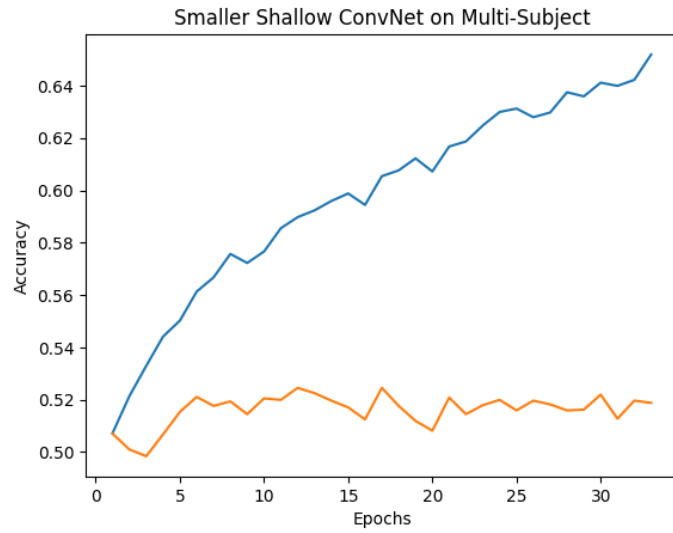


Figure 10: Accuracy across time for Smaller Shallow ConvNet across all subjects.

This attempt to mitigate overfitting did delay overfitting to later epochs, but the validation accuracy still remained low. Therefore, other methods to extract useful patterns are necessary.

7.2 Averaging

To increase the signal to noise, we averaged every 10 trials of the same condition. This led the total number of trials to go from 17,421 to 1,693. Each subject randomly shuffled their trials and then averaged every 10 trials until less than 10 remained and were discarded. With a fewer number of trials to train and test on, we only considered the cross-subject case.

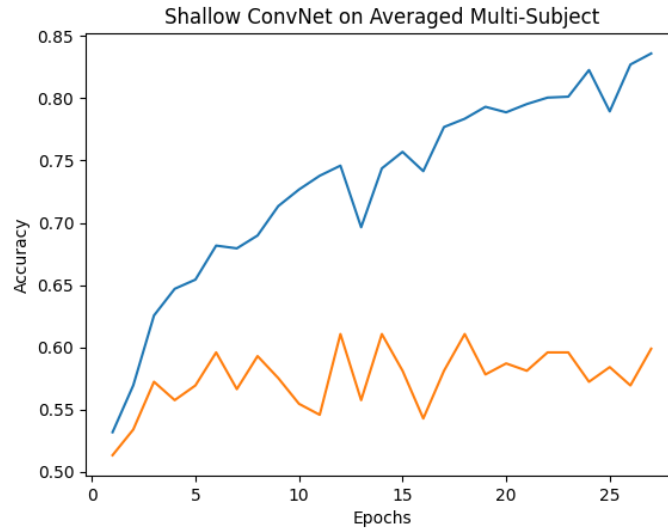


Figure 11: Accuracy across time for Shallow ConvNet across all subjects with averaged trials.

The validation accuracy was able to get above 55%, which previous methods to this were not able to achieve. Furthermore, a the Smaller ConvNet can be used to further reduce overfitting.

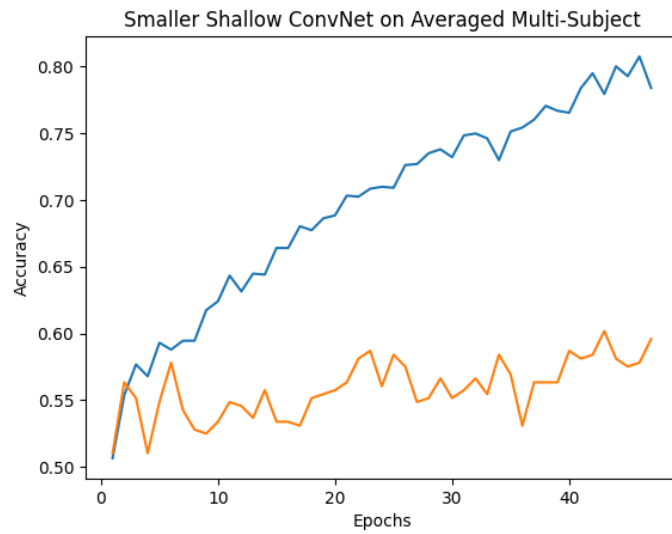


Figure 12: Accuracy across time for Smaller Shallow ConvNet across all subjects with averaged trials.

Given this new modified model with a procedure to increase the signal to noise ratio, the validation accuracy was more consistently able to reach above 55%.

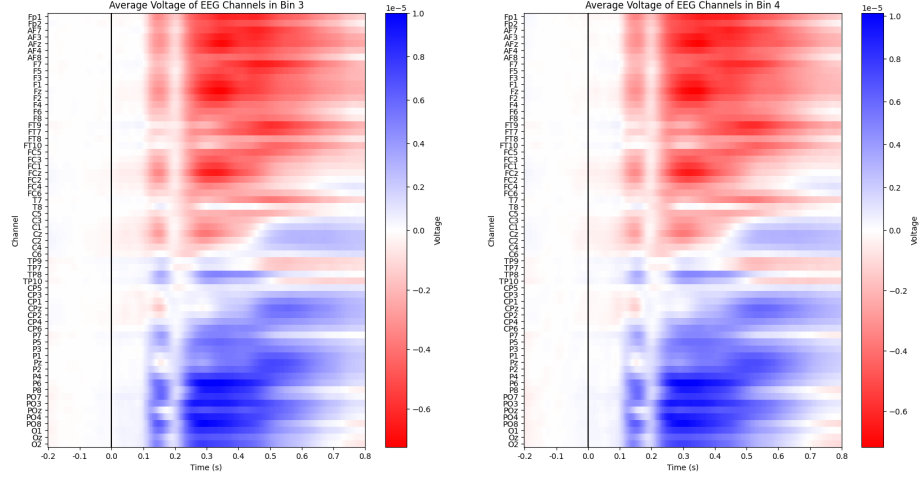


Figure 13: Average voltage from each channel in Bin 3 vs Bin 4. The standard deviation for Bin 3 is $\sim 8.65e-6$ and for Bin 4 is $\sim 8.66e-6$, making it infeasible to decode conditions using channels independently.

The average voltage from each channel in Figure 13 shows increased relevant activity around the brain on average after around 100 milliseconds. However, the difference of voltage between the two conditions is not large enough to either decode the condition or determine which parts of the brain play a larger role in either condition. To detect the earliest time conditions can be decoded and the activations of certain areas during that time, SVMs were used.

Each subject on average has around 300 data points, which is not sufficient for an SVM with 64 features as the model is very likely to overfit. However, consistently across subjects, the validation accuracy stays at chance level (~ 0.5) then increases after around 100 milliseconds after stimulus onset. This result suggests that although the models overfit, the consistent increase at around the same time indicates that there are relevant EEG patterns related to decoding between the two conditions.

Training separate SVMs on individual subjects yields high accuracy, however training an SVM on all subjects is not above chance. Therefore, how the channels are used to decode between the two conditions depends on the individual subject. This makes it difficult to reduce overfitting as each subject requires their own feature selection to determine which channels to remove.